

Ejercicios visuales con while (enunciados + salidas)

La solución muestra exactamente lo que imprime cada programa.

1)

```
r = 3
c = 5
i = 0
while i < r:
    j = 0
    s = ""
    while j < c:
        s = s + "*"
        j = j + 1
    print(s)
    i = i + 1
```

Salida:

```
*****
*****
*****
```

2)

```
n = 5
i = 1
while i <= n:
    j = 0
    s = ""
    while j < i:
        s = s + "*"
        j = j + 1
    print(s)
    i = i + 1
```

Salida:

```
*
**
***
****
*****
```

3)

```
r = 4
c = 6
i = 0
while i < r:
    j = 0
    s = ""
    while j < c:
        if (i + j) % 2 == 0:
            s = s + "*"
        else:
```

```
        s = s + "."
    j = j + 1
print(s)
i = i + 1
```

Salida:

```
*.*.*.
.*.*.*
*.*.*.
.*.*.*
```

4)

```
n = 6
i = 0
while i < n:
    j = 0
    s = ""
    while j < n:
        if j == i:
            s = s + "\\"
        else:
            s = s + "."
        j = j + 1
    print(s)
    i = i + 1
```

Salida:

```
\.....
.\.....
..\.....
...\.....
....\.....
.....\.....
.....\.....
```

5)

```
n = 5
i = 1
while i <= n:
    j = 1
    s = ""
    while j <= i:
        s = s + str(j)
        j = j + 1
    print(s)
    i = i + 1
```

Salida:

```
1
12
123
1234
12345
```

6)

```
r = 4
c = 7
i = 0
while i < r:
    j = 0
    s = ""
    while j < c:
        k = 65 + ((i + j) % 3)
        ch = chr(k)
        s = s + ch
        j = j + 1
    print(s)
    i = i + 1
```

Salida:

ABCABCA
BCABCAB
CABCABC
ABCABCA

7)

```
r = 5
c = 9
i = 0
while i < r:
    j = 0
    s = ""
    while j < c:
        if i == 0 or j == 0 or i == r - 1 or j == c - 1:
            s = s + "#"
        else:
            s = s + " "
        j = j + 1
    print(s)
    i = i + 1
```

Salida:

```
#####
#       #
#       #
#       #
#####
```

8)

```
r = 4
c = 6
i = 0
while i < r:
    j = 0
    s = ""
    while j < c:
        s = s + str(i % 10) + str(j % 10)
        j = j + 1
    print(s)
    i = i + 1
```

Salida:

```
000102030405
101112131415
202122232425
303132333435
```

9)

```
n = 5
i = n
while i > 0:
    j = 0
    s = ""
    while j < i:
        s = s + "*"
        j = j + 1
    print(s)
    i = i - 1
```

Salida:

```
*****
```


**
*

10)

```
r = 5
c = 8
i = 0
while i < r:
    j = 0
    s = ""
    while j < c:
        if (i % 2 == 0 and j % 2 == 0) or (i % 2 == 1 and j % 2 == 1):
            s = s + "*"
        else:
            s = s + "-"
        j = j + 1
    print(s)
    i = i + 1
```

Salida:

```
*_*_*_*_*
_**_**_
*_*_*_*_*
_**_**_
*_*_*_*_
```